

Task 3(hands_on2)

List each student's full name (first_name + ' ' + last_name) alongside the name of their department.
(JOIN students and departments.)

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168 -- task 3
169 -- List each student's full name (first_name + ' ' + last_name) alongside the name of their department. (JOIN students and departments.)
170 • select concat(first_name, ' ', last_name) as full_name, dept_name
171 from students s join departments d
172 on s.department_id = d.department_id;
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full_name	dept_name
Arjun Mehta	Computer Science
Priya Suresh	Computer Science
Vikram Das	Computer Science
Deepika Rao	Computer Science
Sadhana Shree	Computer Science
Rohan Verma	Electronics
Kavya Menon	Electronics
Neha Gupta	Electronics
Sneha Patel	Mechanical
Aditya Singh	Civil

Show each enrollment along with the student's name and the course name. (3-table JOIN: enrollments, students, courses.)

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174 -- Show each enrollment along with the student's name and the course name. (3-table JOIN: enrollments, students, courses.)
175 • select concat(first_name, ' ', last_name) as full_name, course_name, grade, enrollment_year
176 from enrollments e join students s
177 on s.student_id = e.student_id
178 join courses c on c.course_id = e.course_id;
```

full_name	course_name	grade	enrollment_year
Arjun Mehta	Data Structures & Algorithms	A	2022
Priya Suresh	Data Structures & Algorithms	B	2022
Vikram Das	Data Structures & Algorithms	B	2022
Deepika Rao	Data Structures & Algorithms	A	2022
Arjun Mehta	Database Management Systems	B	2022
Vikram Das	Database Management Systems	A	2022
Priya Suresh	Object Oriented Programming	A	2022
Deepika Rao	Object Oriented Programming	B	2022
Rohan Verma	Circuit Theory	A	2021
Kavya Menon	Circuit Theory	B	2021

Find all students who are NOT enrolled in any course using a LEFT JOIN and WHERE ... IS NULL pattern.

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179
180 • select * from enrollments;
181 -- Find all students who are NOT enrolled in any course using a LEFT JOIN and WHERE ... IS NULL pattern.
182 • select *
183 from students s
184 left join enrollments e on e.student_id = s.student_id
185 where e.enrollment_id is null;
186
187
188

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student_id	first_name	last_name	email	date_of_birth	department_id	enrollment_year	enrollment_id	student_id	course_id	enrollment_date	grade
4	Sneha	Patel	sneha.patel@college.edu	2004-01-30	3	2023	NULL	NULL	NULL	NULL	NULL
7	Aditya	Singh	aditya.singh@college.edu	2004-03-22	4	2023	NULL	NULL	NULL	NULL	NULL
11	Sadhana	Shree	sadhana.shree@college.edu	2006-06-17	1	2023	NULL	NULL	NULL	NULL	NULL
12	Neha	Gupta	neha.gupta@college.edu	2004-02-10	2	2023	NULL	NULL	NULL	NULL	NULL

Display every course along with the number of students enrolled in it. Courses with zero enrolments must still appear. (LEFT JOIN courses with enrollments, GROUP BY course.)

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187
188 -- Display every course along with the number of students enrolled in it. Courses with zero enrolments must still appear. (LEFT JOIN
189 SELECT c.course_name,
190        COUNT(e.student_id) AS total_students
191 FROM courses c
192 LEFT JOIN enrollments e
193 ON c.course_id = e.course_id
194 GROUP BY c.course_id, c.course_name;
195

```

course_name	total_students
Data Structures & Algorithms	4
Database Management Systems	2
Object Oriented Programming	2
Circuit Theory	2
Thermodynamics	0

List each department along with its professors and their salaries. Include departments that have no professors yet.

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195
196 -- List each department along with its professors and their salaries. Include departments that have no professors yet.
197 • select dept_name,prof_name,salary
198 from departments d
199 left join professors p on p.department_id = d.department_id
200 order by dept_name;

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dept_name	prof_name	salary
Civil	Dr. Kartik Bose	76000.00
Computer Science	Dr. Anand Krishnan	95000.00
Computer Science	Dr. Meena Pillai	88000.00
Electronics	Dr. Sunil Rajan	82000.00
Mechanical	Dr. Latha Gopal	79000.00